Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Fan Lai	**Email**	fanlai@illinois.edu
Institution / Department	Siebel School of Computing and Data Science			
Address	201 North Goodwin Avenue Urbana, IL			

2. Technology Overview

Title of Proposed Technology	**SLO-Aware Orchestration for AI Services across Heterogeneous Infrastructure**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		Information & Communication	
	Medium Category	*Small Category*	*Medium Category*	*Small Category*
	Automobile/Rail Vehicles	Vehicle Intelligence Technologies	Home Network	Intelligent Home Appliances
Concept of Proposed Technology	Nexus is a software technology that automatically plans and optimizes compound AI pipelines: workflows composed of multiple AI operators (e.g., perception, tracking, retrieval, reasoning, generation, control), across heterogeneous edge–cloud infrastructure. Given application constraints (latency, accuracy, cost, privacy), Nexus selects (i) operator configurations (e.g., model size, precision, resolution), (ii) placements (e.g., on-device, near-edge, private server, or cloud), and (iii) resource allocations (e.g., GPU, CPU, memory, and network budgets) to maximize SLO goodput: the number of requests that meet service-level objectives under real-world contention. Nexus combines cost-aware multi-objective plan search, adaptive accuracy/latency profiling, and contention-aware multi-request scheduling to deliver second-level planning speed and stable service quality. The technology is applicable to mobility stacks (e.g., ADAS/IVI perception + LLM assistant + safety filters), robotics and manufacturing (e.g., vision + inspection + planning), and smart devices where workloads are bursty and environments are dynamic.			

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Rationale for Joint R&D Project	Korean SMEs are world-class at developing specialized AI hardware (NPUs) and vertical-specific applications (e.g., medical imaging, factory automation). However, they often face the "Edge-AI Gap": the need to deploy multi-model stacks on embedded platforms with strict safety and latency requirements. Current manual tuning is unscalable and brittle. A joint R&D effort is essential to bridge UIUC's systems-level algorithmic leadership with the domain-specific hardware and production data of a Korean SME. UIUC provides the "Nexus" brain, the core planning algorithms and adaptive profiling engines. The SME partner provides the "Body", the proprietary sensor modalities, in-vehicle/factory compute platforms, and real-world safety-critical workloads. Together, we will co-develop an integrated serving stack that allows the SME to move from cloud-only deployments to pervasive AI, providing a significant competitive advantage in the global SDV and robotics markets.
Keywords	1. Edge-Cloud AI 2. Compound AI Serving 3. SLO-Aware Systems 4. Multi-Model Pipelines 5. Collaborative AI

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)

R&D Objectives

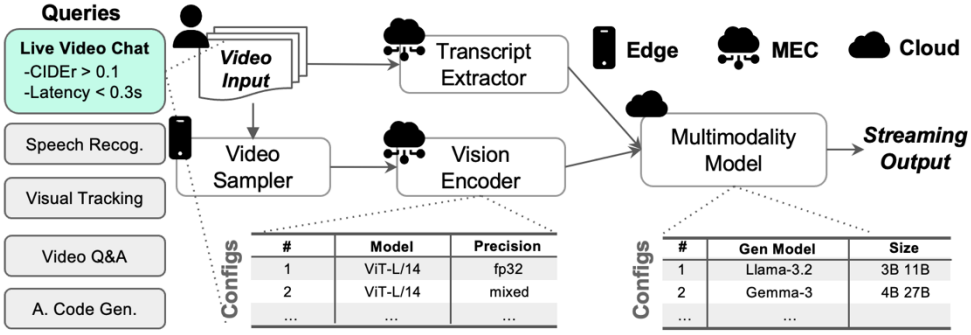
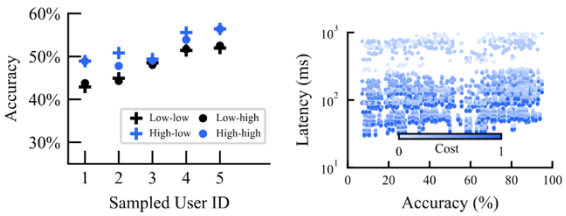


Figure 1: As AI services increasingly cater to the end user, their deployments can span edge-cloud infrastructure.

The rise of compound AI serving that integrates multiple operators in a pipeline across edge and cloud infrastructures is enabling transformative end-user experiences (as shown in Figure 1), from generative AI-assisted asset creation to immersive gaming with intelligent NPCs, and real-time scene analysis for vision sensing. Achieving high service goodput--i.e., meeting service level objectives (SLOs) for pipeline latency, accuracy, and costs--requires joint planning of operators' placement, configuration, and resource allocation. However, diverse SLOs, varying runtime environments (e.g., device speeds), and intense multi-query contention explode the planning space (Figure 2), making real-time serving and cost-efficient deployment intractable with existing advances [1, 2].



(a) Heterogeneous Data. (b) Vast Search Space.

System	Support	Respon.	Near-Opt.
Exhaust. Search	✓, ✓, ✓, ✓	> 2 hrs	> 24 hrs
VStorm [37]	✓, ×, ×, ✓	~1.6 min	~34 min
Vulcan [39]	×, ✓, ✓, ×	~1.2 min	~7 min
Our Target	✓, ✓, ✓, ✓	*< 5s*	*< 20s*

(c) Existing query planners take minutes to hours to find the first SLO-compliant plan (``Responsive.") and near-optimal plans (``Near-Opt."), with limited ``Support" for [multi-query, multi-tier, heterogeneous machines, contention-awareness], hurting user-facing services.

Figure 2: (a) Input distributions vary across users, causing accuracy variance for the same plan. ``Low-high": resource-light configurations for operator 1 and resource-intensive configurations for operator 2. (b) Each query can have thousands of plans. (c) Existing planners face fundamental limitations for practical deployments.

This proposal plans to develop Nexus, the first SLO-aware query planner optimized for large-scale compound AI workloads across edge-to-cloud deployments. At its core is to decompose the many-query, multi-SLO planning problem into tractable subproblems while preserving global decision quality, exploiting plan similarities within and across queries to slash the search steps. At runtime, Nexus will perform query-plan bipartite matching to maximize SLO goodput under resource contentions. Nexus will substantially elevate user experiences, reduce operational costs, and maximize infrastructure utilization, enabling service providers to deliver next-generation AI-powered applications to millions of end users seamlessly.

Current capabilities (available now):

- Prototype formulation of compound AI planning as an SLO-constrained optimization over configuration/placement/resources.

	• Initial guided search for SLO-feasible pipeline candidates and basic contention modeling. • Early profiling harness for operator-level latency/accuracy measurement and replay-driven evaluation. To be developed via this joint R&D: • Cost-aware multi-objective planner that finds SLO-feasible pipeline plans in seconds, not minutes. • Adaptive profiling that reduces GPU profiling time while preserving correctness of SLO feasibility decisions. • Dependency-aware caching that reuses shared operator measurements across plan variants and across queries. • Contention-aware multi-request scheduler that selects query–plan combinations to maximize goodput on shared resources. • Online replanning triggered by network/device/environment drift, with bounded overhead and minimal service disruption. • Integration package (APIs + deployment recipes) for the SME's serving stack on edge devices and servers.
R&D Target Specifications	Nexus will target measurable system-level outcomes on partner-selected pipelines and platforms: Planning latency: • Time-to-first SLO-feasible plan: ≤ 5 seconds (target) • Time-to-near-optimal plan set (Pareto candidates): ≤ 20 seconds (target) Profiling cost reduction: • Reduce accuracy/latency profiling compute by ≥ 5× versus fixed-sample profiling baselines (target) • Reuse/caching hit rate: ≥ 50% operator-profile reuse across plan variants (target, workload-dependent) Service goodput improvement under contention: • Increase fraction of requests meeting end-to-end SLOs by ≥ 20–40% compared to per-query independent planning (target range) Cost / utilization: • Reduce total accelerator cost per served request by ≥ 15–30% while preserving SLOs (target range) • Improve accelerator utilization at equal SLO compliance by ≥ 10–20% (target range) Robust adaptation: • Detect and replan within ≤ 30 seconds after significant bandwidth/compute shifts; restore SLO compliance for affected traffic within ≤ 2 minutes (target)
R&D Plan	Phase 1 (0–6 months): Seed-stage planning and prototype integration plan • Task 1: Select 2–3 representative partner pipelines (e.g., perception → tracking → decision; RAG/LLM assistant with safety; visual inspection). • Task 2: Build the measurement harness (operator profiling + end-to-end tracing) on partner platforms. • Task 3: Implement Nexus v0: single-query cost-aware multi-objective search + baseline contention model. • Deliverables: pipeline specs; profiling dataset and replay tool; Nexus v0 demo; Phase 2 full plan. Phase 2 (6–18 months): Core algorithms + system prototype • Task 4: Adaptive profiling with statistical early stopping; correctness criteria for SLO feasibility. • Task 5: Dependency-aware caching across operator variants; reuse rules across placements/configs. • Task 6: Multi-query planner: admission + plan selection + placement under resource constraints. • Deliverables: Nexus v1 with APIs; experimental report on partner workloads; integration with partner

	staging environment. Phase 3 (18–30 months): Reliability, productization, and field validation • Task 7: Online adaptation policies for bandwidth/device variability; safe replanning without service disruption. • Task 8: Production-like validation: long-running tests, failure modes, regression checks, and KPI tracking. • Task 9: Deployment toolkit: packaging, monitoring hooks, and operator onboarding guide for new pipelines. • Deliverables: Nexus v2 pilot; validation results; documentation; commercialization transition plan with partner.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Design and implementation of the core optimization algorithms: • Develop the Nexus planning algorithms (multi-objective search, adaptive profiling, caching, multi-query scheduling). • Build and maintain the prototype software stack, including instrumentation, profiling harness, and planner APIs. • Conduct controlled experiments and root-cause analysis for SLO violations and performance regressions. • Provide technical documentation, integration guidelines, and reference deployment configurations. • Support IP strategy consistent with program requirements; ensure export-control compliant technical disclosures.
Expected R&R of Korean SME Partner	Provision of domain-specific data and real-world AI models (the "workload"): • Provide target use cases, pipeline definitions, and success metrics (latency/quality/cost/privacy constraints). • Provide access to representative platforms (edge devices/servers), runtime environments, and test datasets (as permissible). • Implement or expose necessary hooks in the existing serving stack (telemetry, resource controls, deployment interface). • Lead product-facing validation: QA processes, reliability tests, and (when possible) pilot deployments with users/customers. • Assign engineering resources to co-develop integration, troubleshoot, and iterate on deployment constraints.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D will deliver a deployable planning layer that makes compound AI pipelines reliable and cost-efficient on heterogeneous edge–cloud infrastructure. • Technological Achievement: Establishing a world-leading software framework for Compound AI that surpasses current state-of-the-art systems in planning speed and cost-awareness. • Commercialization Potential: The SME partner can market a "System-Software-Hardware" bundle that is significantly more efficient than competitors using generic MLOps tools. This is directly applicable to Smart Cities, Autonomous Mobility, and GenAI-as-a-service. • Economic Impact: Reducing the "barrier to entry" for Korean SMEs to deploy expensive LLMs and Vision models by optimizing resource usage, potentially saving millions in server costs. The project also establishes a reusable foundation: as new models/operators appear, the partner can onboard them into Nexus rather than redesigning deployment rules from scratch.

6. References

(1) Yiwen Zhang, Xumiao Zhang, Ganesh Ananthanarayanan, Anand Iyer, Yuanchao Shu, Victor Bahl, Z Morley Mao, and Mosharaf Chowdhury. 2024. Vulcan: Automatic Query Planning for Live ML Analytics. NSDI, 2024.

(2) Haoyu Zhang, Ganesh Ananthanarayanan, Peter Bodik, Matthai Philipose, Paramvir Bahl, and Michael J Freedman. 2017. Live video analytics at scale with approximation and Delay-Tolerance. NSDI, 2017.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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